RISK 201: Bonds

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6.1 Introduction and terminology

- Bonds are financial instruments that corporations and governments use to raise capital
- They are debt instruments:
  - The bond issuer is the borrower: gets cash at issue, and will pay back principal and interest
  - The bondholder (investor) is the lender
- If the bond issuer does not make a payment as scheduled, we say they defaulted on the bond
- Bonds are rated based on their probability of default

Bond ratings

Bond Rating			
Moody's	**Standard & Poor's**	**Grade**	**Risk**
Aaa	AAA	Investment	Lowest Risk
Aa	AA	Investment	Low Risk
A	A	Investment	Low Risk
Baa	BBB	Investment	Medium Risk
Ba, B	BB, B	Junk	High Risk
Caa/Ca/C	CCC/CC/C	Junk	Highest Risk
C	D	Junk	In Default

Source: Investopedia (image by Sabrina Jiang © Investopedia 2020)

Types of government bonds

- **Savings bonds**: redeemable any time (issuer will take it back and pay principal plus interest)
- **Marketable bonds**:
  - Have a specific maturity date
  - Usually not redeemable before maturity
  - Bondholder can sell it to another investor
  - Market price will depend on the characteristics of the bond as well as market conditions at time of sale

Terminology

Face value (par value)	Amount noted on front (face) of bond. Usually multiples of 100.
Maturity date	The date on which the loan is (normally) repaid.
Callable bond	A bond that can be redeemed before maturity at the discretion of the bond issuer, at specific dates, for specific amounts
Redemption date	A date on which the loan can be repaid
Redemption value	Amount payable at redemption date. "At par" means at face value.

Terminology, continued

Coupon rate (bond rate)	The rate at which the bond pays interest on its face value until the maturity date. Coupon payments are made at equal intervals (usually semiannually).
Yield rate (investment rate)	The rate of return realized by the bondholder on its investment.

Notation

F: face value

C: redemption value

r: coupon rate per interest period

i: yield rate per interest period

n: number of interest periods until maturity date

P: purchase price of the bond

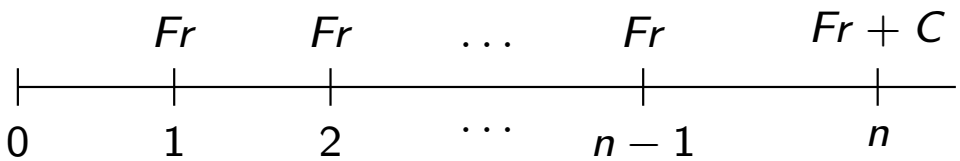
Fr: coupon payment, received at the end of each interest period

"Yield"

- In practice, the coupon rate and the yield are quoted as nominal rates convertible semi-annually. For example, a "5% coupon bond yielding 8%" has $r =$ and $i =$.
- Unless stated otherwise, the "yield" we will be calculating is the yield to maturity (YTM).
- If the bond is sold before maturity, the investor may realize a different yield.

6.2 Purchase price of a bond

- Assumptions:
  - No default
  - Fixed maturity date
- Price is the present value of future payments (coupons plus redemption value)



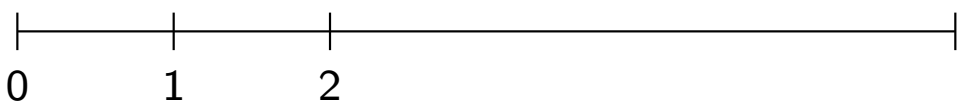
$$P = Fr\, a_{\overline{n}|i} + Cv^n$$

Example 6.1

A \$1000 face value bond with 6% coupons is redeemable at \$1200 at the end of 5 years.

(a) What is the purchase price that yields 5%?

$F =$ $C =$ $n =$ $r =$ $i =$

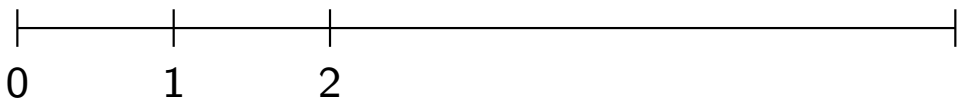


$P =$

Example 6.1, continued

(b) What is the purchase price that yields 8%?

$F =$ $C =$ $n =$ $r =$ $i =$

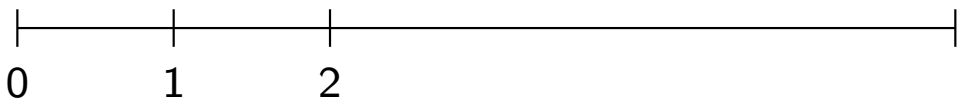


$P =$

Example 6.1, continued

(c) What is the purchase price that yields 4%?

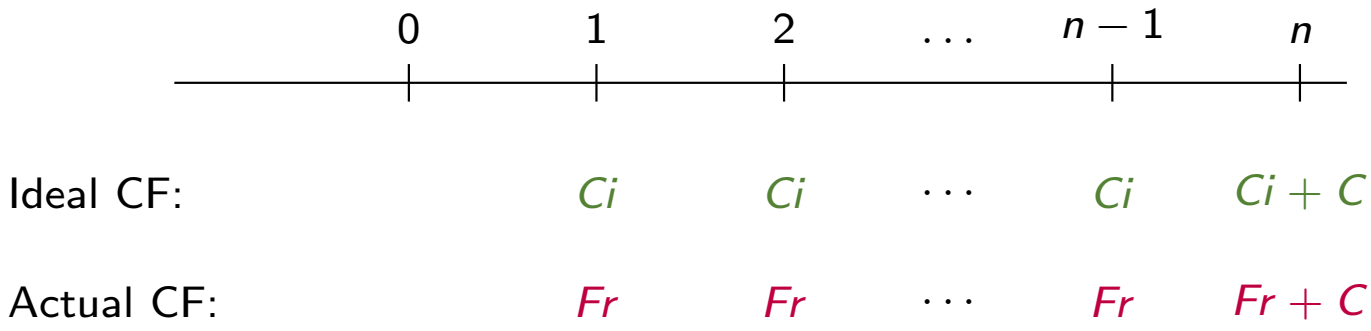
$F =$ $C =$ $n =$ $r =$ $i =$



$P =$

Premium/discount formula

An alternative formula for the price of a bond



$$P = C + (Fr - Ci)\, a_{\overline{n}|i}$$

Example 6.1, continued

Confirm the prices calculated earlier using the premium/discount formula

$$F = \qquad C = \qquad n = \qquad r =$$

(a) $i = 2.5\%$

(b) $i = 4\%$

(c) $i = 2\%$

Auction process for GoC bonds

- Gov't decides on face amount, term, coupon rate
- Invites bids from investors for the yield they are willing to accept (to 3 decimals), and the amount of face value they are willing to buy
- Yield is converted to price (5 decimals accuracy)
- "First price sealed bid" auction:
  - bids ranked from lowest to highest in terms of yield
  - bids accepted in rank-order until desired amount is raised
  - bids lower than threshold not accepted
  - all accepted bids get the yield they proposed

Auction process, continued

- Different from Dutch auction
- Who can bid, and for what quantity, is limited
- Competitive vs. non-competitive bids

Notes:

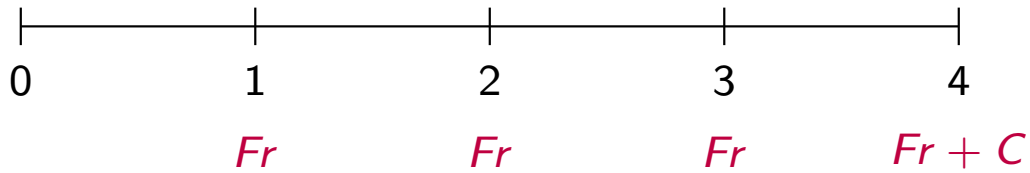
- Process for issuing corporate bonds is different
- After issue, bonds are traded in a dealer resale market

6.3 Premium and Discount

- Recall that a bond is essentially a loan from the investor to the bond issuer.
  - The amount of the loan is the purchase price P
  - The periodic payments Fr together with the redemption value C are enough to pay back the loan at interest rate i per period
- We can draw up an amortization schedule for this loan, just like in Chapter 5

Example 6.2

Consider a \$1000 face value bond with 6% coupons, redeemable for \$1200 at the end of 2 years, yielding 5%.



$$P = C + (Fr - Ci)\, a_{\overline{n}|i} =$$

Example 6.2, continued

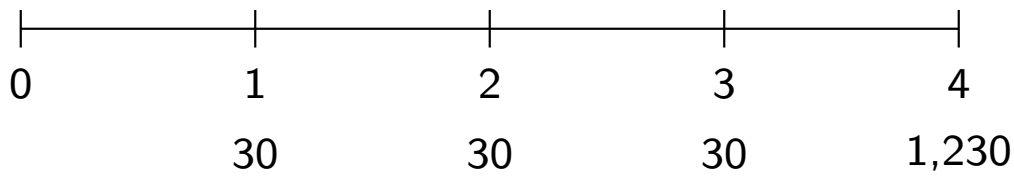
Period	**Payment**	**Interest (at i)**	**Principal**	**Outstanding balance**
0				1,200
1	30			
2	30			
3	30			
4	30			
4	1,200			

Observations

- When $P = C$, the coupon payments (Fr) exactly cover the interest accruing each period and no principal is paid back until maturity.
- The outstanding balance on the loan is known as the book value of the bond.
- The book value immediately after a coupon payment can be calculated as the PV of the remaining payments (coupons plus redemption value) at rate i per period. You can use either pricing formula to find this value.

Example 6.3

Consider a \$1000 face value bond with 6% coupons, redeemable for \$1200 at the end of 2 years, yielding 3.1%.



$$P = C + (Fr - Ci)\, a_{\overline{n}|i} = 1{,}243.89$$

$P > C$ we say the bond is purchased at a premium

Example 6.3, continued

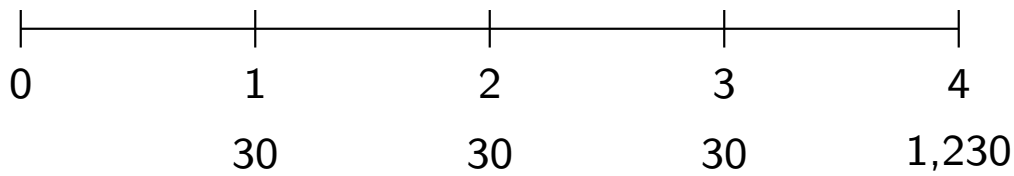
Period	Payment	Interest (1.55% per period)	Principal	Outstanding balance
0				1,243.89
1	30.00			
2	30.00			
3	30.00			
4	30.00			
4	1,200.00			

Observations ($P > C$)

- When $P > C$, a portion of each coupon (Fr) goes towards repaying principal.
- For a bond purchased at a premium, the book value gradually decreases from P at time 0 to C just after the last coupon payment. We say the book value is being written down, or the premium is amortized.
- The amortization of the premium in each coupon payment is the principal repayment portion.
- Successive book value adjustments are in ratio $1 + i$.
- The total amortization of premium is $P - C$.

Example 6.4

Consider a \$1000 face value bond with 6% coupons, redeemable for \$1200 at the end of 2 years, yielding 8%.



$$P = C + (Fr - Ci)\, a_{\overline{n}|i} = 1{,}134.66$$

$P < C$ we say the bond is purchased at a discount

Example 6.4, continued

Period	Payment	Interest (4% per period)	Principal	Outstanding balance
0				1,134.66
1	30.00	45.39	−15.39	1,150.05
2	30.00	46.00	−16.00	1,166.05
3	30.00	46.64	−16.64	1,182.69
4	30.00	47.31	−17.31	1,200.00
4	1,200.00		1,200.00	0.00

Observations ($P < C$)

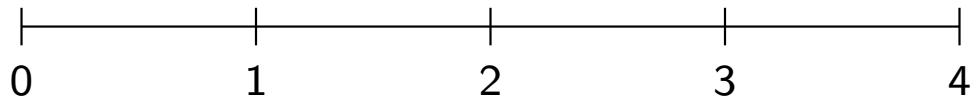
- When $P < C$, the coupons are not enough to cover the interest accruing each period so the book value increases over the life of the bond. We say the book value is being written up, or the discount is accumulated.
- The amount for accumulation of discount in each coupon payment is the amount by which the principal increases.
- Successive book value adjustments are in ratio $1 + i$.
- Total book value adjustment is $P - C$.

Book value vs. market value

- The **book value** of a bond is the present value of the remaining coupons plus C at the yield rate (i) in effect at the time of purchase.
- After purchase, the book value may not be the price for which the bond could actually be sold in the open market.
- If the yield rates available in the market at a later date (i_t) are different from the rate i available at the time of purchase, then the **market value** of the bond will be different from the book value.

Example 6.5

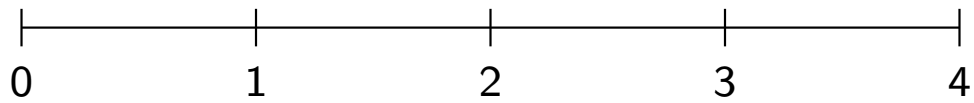
Consider the bond in Example 6.4 ($F = 1000$, $C = 1200$, $r = 3\%$, $i = 4\%$). The book value at the end of the first year is \$1,166.05.



Find the **market value** at the end of the first year if the prevailing yield rate is 12% convertible semi-annually.

Example 6.5, continued

If the investor bought the bond for \$1,134.66 at issue and sold it at market value at the end of the first year, what is the realized yield over the 1-year holding period?



6.4 Price between coupon dates

- Purchase price formulas from 6.2 only work when the bond is issued or immediately after a coupon payment, so the remaining coupons are equally spaced.
- If a bond is purchased between coupon payment dates, we need to adjust the price.
- The book value of the bond increases as we get closer to each coupon payment date, and then drops when the coupon is actually paid.

Full price of a bond

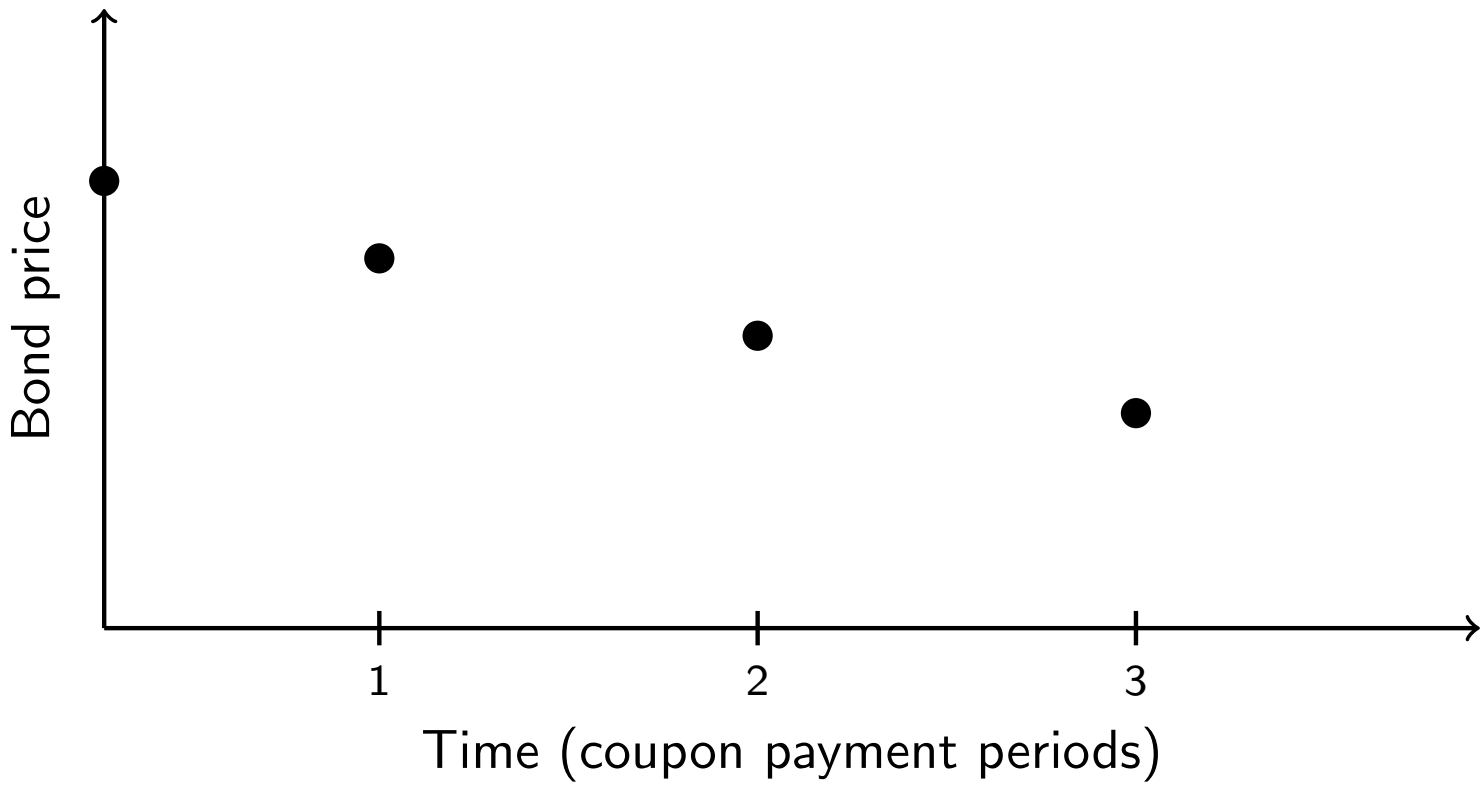
- The **full price of a bond** is the total value of all future payments as of the purchase date.
- The full price includes all of the value of the next coupon payment
- At a point in time that is a portion $0 < k < 1$ of the way between two coupon payment dates, the full price is

$$P = P_0(1 + i)^k$$

- At the coupon payment date, the full price drops to

$$P_1 = P_0(1 + i) - Fr$$

Evolution of full price



6.6 Determining the yield rate

To determine the yield on a bond:

- Use a financial calculator
- Use Excel (YIELD)
- Use interpolation between integer values
  - Needs a starting point
  - Consider the "method of averages" which approximates the true yield

The method of averages

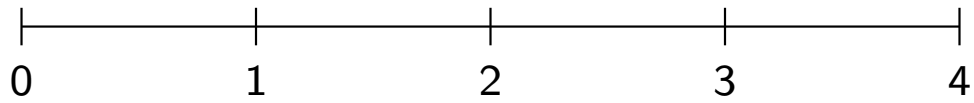
$$\text{Approximate yield} = \frac{\text{Average interest payment}}{\text{Average amount invested}}$$

$$\text{Average interest payment:} \quad \frac{n \cdot Fr + C - P}{n}$$

$$\text{Average amount invested:} \quad \frac{P + C}{2}$$

Example 6.6

Consider a \$1000 face value bond with 6% coupons, redeemable for \$1200 at the end of 10 years. If the bond sells for \$1054, what is the yield?



Average interest:

Average amount invested:

Approximate yield:

6.7 Other types of bonds

A **zero-coupon bond** (aka zero)

- Makes no coupon payments
- Pays the face value at maturity
- Sold at a discount relative to face value
  - Aka discount bond or pure discount bond
- Price is PV of face value, discounted at yield rate
- Examples:
  - T-bills (short term, maturity < 1 year)
  - T-notes (intermediate term, less than 5 years)
  - Stripped bonds (longer term)

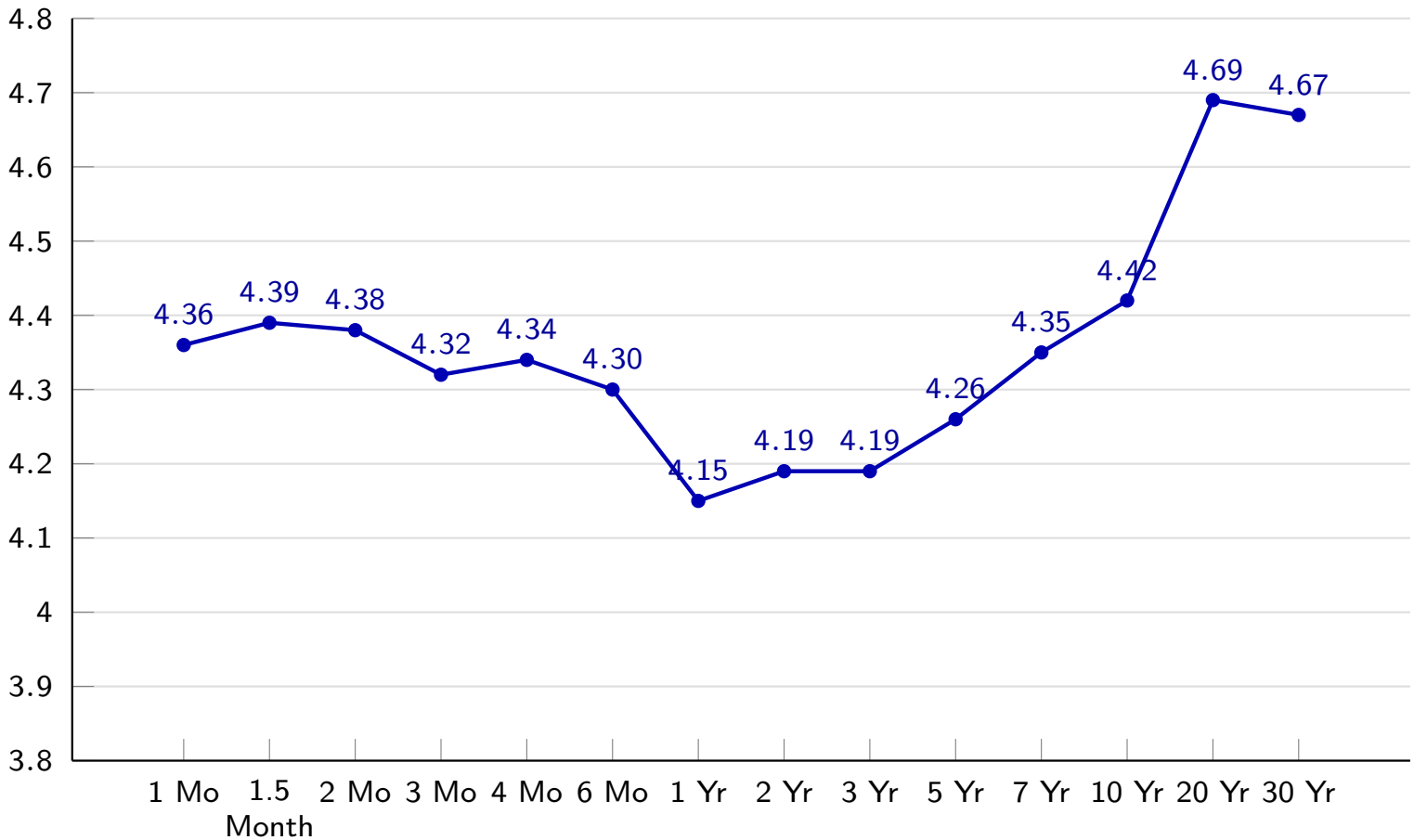
Stripped bonds

- An investor could:
  - Buy a regular bond with coupons
  - Resell the coupon payments and the redemption value separately
- The "redemption only" part is known as a **strip bond**
- Coupon payments could be sold individually or in groups

Stripped bonds, continued

- In the US, the Separate Trading of Registered Interest and Principal of Securities (STRIPS) program started in 1985.
- In Canada, Canadian Depository Services Ltd. (CDS) has been stripping bonds electronically since 1987.

Observed US Treasury yields (Feb 21)



6.8 Callable Bonds

- A callable bond is a bond which gives the issuer (not the bondholder) the right to redeem prior to its maturity date, under certain conditions.
- When issued, the **call provisions** explain when the bond can be redeemed and what the price will be.
- In most cases, there is **some period of time** during which the **bond cannot be called**. This period of time is named the **call protection period**.

Callable Bonds, continued

- The earliest time to call the bond is named the **call date**.
- The **call price** (could be different from the redemption value) is the amount of money the issuer must pay to buy the bond back.
- Usually, bonds can be only called immediately after the payment of a coupon. If the bond is called immediately after the payment of k-th coupon, the present value of the obtained payments is

$$P_k = C + (Fr - Ci)\, a_{\overline{k}|i}$$

Callable bond pricing

- P_k is the price which the bondholder would pay for the bond assuming that the bond is called immediately after the k-th coupon. The smaller P_k is, the worse it is for the bondholder.
- Between all possible choices to recall a bond, the issuer will choose the option with the **smallest price**.

Callable bond pricing

(i) if $Fr > Ci$ (bond sells at a premium), P_k increases with k, and we assume the **redemption date is the earliest possible**.

(ii) if $Fr < Ci$ (bond sells at a discount), P_k decreases with k, and we assume that the **redemption date is the latest possible**.

- If a bondholder wants to get an effective rate of interest of i per period, then the maximum price which the bondholder should pay is the lowest possible P_k under that particular rate i.

Example 6.7

Consider a bond with $F = 100$, $r = 8\%$ with annual coupons callable at

- 110 starting 10 years after issue for the next 5 year,

- and maturing at 115 at the end of 15 years.

What is the highest price which a bondholder can pay and still be certain of an annual yield of 7%?

Example 6.7, continued

Case 1) If the bond is redeemed when $10 \leq k < 15$:

We have $C = 110$ and $Fr - Ci = 0.3 > 0$ so that the bond is sold at a premium (i.e., P_k is increasing with k) so that it will be redeemed the earliest possible. Thus, the call price with Case 1 is

$$P_{10} = 0.3\, a_{\overline{10}|0.07} + 110 = 112.11.$$

Example 6.7, continued

Case 2) If the bond is redeemed when $k = 15$:

We have $C = 115$ and $Fr - Ci = -0.05 < 0$ so that the bond is sold at a discount (i.e., P_k is decreasing with k) so that it will be redeemed the latest possible. Thus, the call price with Case 2 is

$$P_{15} = -0.05\, a_{\overline{15}|0.07} + 115 = 114.54.$$

Therefore, the price of the callable bond is

$$\min(112.11,\ 114.54) = 112.11.$$

SOA FM Samples

https://www.soa.org/4a5549/globalassets/assets/files/edu/2018/2018-10-exam-fm-sample-questions.pdf

72, 135, 136, 139, 142, 198, 292, 301, 378, 381